

Final Year Project Proposal

Project Title: Design and Implementation of a Cross-Platform Dissertation Record System

Degree: Bachelor of Science with Honours in Computer Science (BSc Hons)

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Institution: Njala University

1. Introduction & Background

In modern academic institutions, managing final year dissertations—from topic submission and supervisor allocation to milestone tracking and feedback—is often plagued by fragmented communication and manual record-keeping. To bridge this gap, this project proposes a comprehensive, secure, and cross-platform **Dissertation Record System**. The system will streamline the academic research lifecycle, providing a centralized repository for students, supervisors, and department heads to interact, submit deliverables, and track progress transparently.

2. Problem Statement

The current process of managing dissertation milestones relies heavily on physical documentation, unstructured emails, and manual tracking. This leads to several inefficiencies:

- **Lack of Progress Transparency:** Students and supervisors lack a unified visual timeline to track chapter submissions and approvals.
- **Data Fragmentation:** Feedback, grading metrics, and draft versions are scattered across different platforms, risking loss of data integrity.
- **Limited Access:** Stakeholders restricted to desktop environments face delays in critical communication, highlighting the need for an integrated mobile solution.

3. Aim and Objectives

The primary aim of this project is to develop a robust, integrated web and mobile ecosystem that centralizes and secures the dissertation management workflow.

Specific Objectives:

1. To design and implement a secure, relational database using **SQL Server** to store user credentials, dissertation milestones, and supervisor feedback.
2. To develop a scalable, secure **RESTful API** using **Visual Studio 2022** and **ASP.NET Core**, documented via **Swagger**, to handle the system's core business logic.
3. To construct a responsive web frontend using **HTML5** and **CSS3** via **Visual Studio Code** for comprehensive administrative and desktop management.
4. To develop a cross-platform mobile client using **Flutter** and **Dart** to provide real-time updates, notifications, and dashboard tracking for users on the move.

4. Methodology & Technical Architecture

This project follows the Full-Stack Software Engineering paradigm, employing a **Three-Tier Architecture** to ensure clean separation of concerns and high scalability:

[Presentation Layer] --> [Logic Layer] --> [Data Layer]

- Web Client (HTML/CSS) - ASP.NET API - SQL Server
- Mobile Client (Flutter/Dart) (Swagger UI) Database

Technical Stack:

- **Data Tier: SQL Server** — Handles relational storage, enforcing data integrity and constraint checks for academic records.
- **Logic Tier (Backend): Visual Studio 2022 / ASP.NET Core API** — Exposes RESTful endpoints. **Swagger** is integrated directly to provide an interactive API blueprint, enabling seamless data binding for the frontends.
- **Presentation Tier (Frontends):**
 - *Web Client:* Built from scratch in **VS Code** using native **HTML5** and **CSS3**, focusing on responsive layout structures (Flexbox/Grid) and semantic web accessibility.
 - *Mobile Client:* Built using **Flutter** and **Dart**, consuming the central API via asynchronous HTTP requests to bring real-time dashboards to mobile devices.

5. Scope of the System (Key Features)

- **Role-Based Access Control (RBAC):** Secure, separate interfaces and permissions for Students, Supervisors, and Department Administrators.
- **Milestone & Progress Tracking:** Visual roadmap indicators showing dissertation progress (e.g., Proposal $\rightarrow$ Methodology $\rightarrow$ Final Draft).
- **Centralized Feedback Loop:** A structured repository where supervisors can leave time-stamped feedback on specific submissions.
- **Cross-Platform Synchronization:** A uniform user experience where actions taken on the web interface instantly reflect on the Flutter mobile app via the shared database.

6. Expected Outcomes and Significance

Upon completion, this project will provide Njala University's Computer Science department with a production-ready application that enhances operational efficiency. For the candidate, this project demonstrates Honours-level mastery over the full software development lifecycle—encompassing relational database design, backend API engineering, semantic web practices, and modern cross-platform mobile development.

Submitted by: Francis Patrick Mukeh Kopoi

Date: June 24, 2026